

Never Forget

Most of us don't pay much attention to the RAM we buy – big mistake!

Memory is one component that has seen an alarming surge in its use over the past two years. Applications have suddenly become memory intensive and a couple of years back you could get along fine with 512 MB of RAM; these days we're seeing even PCs equipped with 1 GB of RAM staggering to meet requirements. Inside your PC, RAM is one of the most important components besides your processor and graphics solution. In fact in many cases RAM is as important as your CPU, simply because the fastest CPU coupled with a meagre amount of RAM will end by bottlenecking performance.

For a reasonably mid-range computer, a simple upgrade to the memory subsystem, say an upping of system RAM from 1 GB to 2 GB, could see an increase in performance of more than 40 per cent in some cases, although this largely depends on the application being used. If you already have a Core 2 Duo processor in the speed range of 2.4 GHz and one GB of RAM, you are better off upgrading to two GB of RAM, rather than changing over to a 3.0 GHz Core 2 Duo. Obviously you would get a performance hike in both cases but in the case of upgrading your RAM this would be more pronounced.

DDR SD RAM is history and has been for some time. DDR2 memory is the norm these days and prices have dropped dramatically. In fact DDR2 RAM is so cheap that you can pick up a four GB kit for a ridiculous amount. Earlier DDR2 533 MHz and 667 MHz was the norm; but you would be hard pressed to find either in the market today. DDR2 800 MHz memory is what is the norm for desktops today and this is widely available. DDR2 1066 MHz memory is still reserved for enthusiasts, although two and four GB kits are available. A two GB kit of DDR2 1066 MHz memory is available for Rs 8,000 to Rs 10,000 from brands like Corsair, Kingston, OCZ and Transcend. 1 GB of DDR2 800 MHz memory costs approxi-

mately Rs 1,000, which means a 4 GB, dual channel kit would cost you Rs 4,000. You may pay a little extra for higher density 2 GB sticks, and a 2 x 2 GB, DDR2 800 MHz configuration would come to Rs 4,500. The four above mentioned brands are the biggest players in India, although brands like GSkill and AData are also available from certain vendors.

From the prices it would seem now's the time to swoop in and swallow a four GB kit. Normally it would be, but there's one problem looming ahead; Intel, with its considerable sway in the way of things with desktop computers has decided to move to DDR3 for its new platform based around the upcoming Nehalem processors. This is an important consideration even if you

do not plan to adopt Nehalem straightaway because this is exactly what happened to DDR memory. Intel decided that DDR2 would benefit performance on Intel based systems more than DDR, and despite AMD's protests, DDR had to go. As of now, both the latest Phenom and Yorkfield processors from AMD and Intel (respectively) support DDR2 but if Nehalem picks up sales within the next six months, the price of DDR2 could rise with it; owing to scarcity as manufacturers stop making it.

Now you're faced with an interesting dilemma. Do you stock up on DDR2 now, in the hope that your PC will last for another two to three years without needing DDR3? Or do you spend big and invest in DDR3 right now, and wait for Nehalem platforms and CPUs to become more affordable? DDR3 is costly right now; and a 4 GB kit of 1,333 MHz DDR3 could easily set you back by Rs 15,000 or so. Faster kits in the range of 1,600 MHz are available from Corsair, OCZ and Kingston, but such kits cost in excess of Rs 20,000. With the way memory is being used, it also doesn't make sense to buy 2 GB anymore, as 64-bit operating systems are the future and these will need at least 4 GB. Also DDR3 shows performance increments over fast DDR2 at speeds of 1,600 MHz and greater; this is because DDR3 is plagued with higher latencies at lower speeds.

My advice is to wait. If you have a Core 2 Duo, or AMD Phenom based PC with 2 GB of memory, hold your horses for another six months. Samsung, Hynix and Qimonda have promised DDR3 2000 and DDR3 2200 MHz memory, which will mean slower clocked memory will become cheaper next year. Do not buy 4 GB of DDR2 today, unless you're sure you will not adopt Nehalem or any other new DDR3 based platform for another 18 months. Only if you absolutely cannot wait and have a budget should you buy DDR3 now. Six months down the line, prices may halve and make your costly investment seem worthless. ☐

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